

# 40106 Glitch Clock

Build an adjustable Schmitt-trigger RC oscillator with a protected line-level pulse output.

## EVIDENCE STATUS

Calculated circuit relationships + browser-simulated audio. Construction layout is diagram-audited but NOT physically tested.

## 1. Project purpose

Build an adjustable Schmitt-trigger RC oscillator with a protected line-level pulse output.

### How it works

The capacitor voltage crosses the inverter's two switching thresholds. Hysteresis flips the output at different rising and falling voltages, so the resistor charges the capacitor in alternating directions and creates oscillation.

### Calculated relationship

**Formula:** Classroom estimate:  $f = 1/(2.2 R C)$

**Substitution:**  $R = 47 \text{ k}\Omega$ ,  $C = 0.1 \text{ }\mu\text{F}$

**Result:** Calculated estimate: 96.7 Hz. Actual frequency depends strongly on CD40106 thresholds, supply, and tolerances.

### Safety boundary

Use only a regulated, current-limited 9 V DC supply. Disconnect power before moving parts. Check IC orientation, diode bands, electrolytic polarity, and rail continuity. Connect the output only to a powered speaker, amplifier, or audio-interface input - never directly to a bare low-impedance speaker.

## 2. Required components

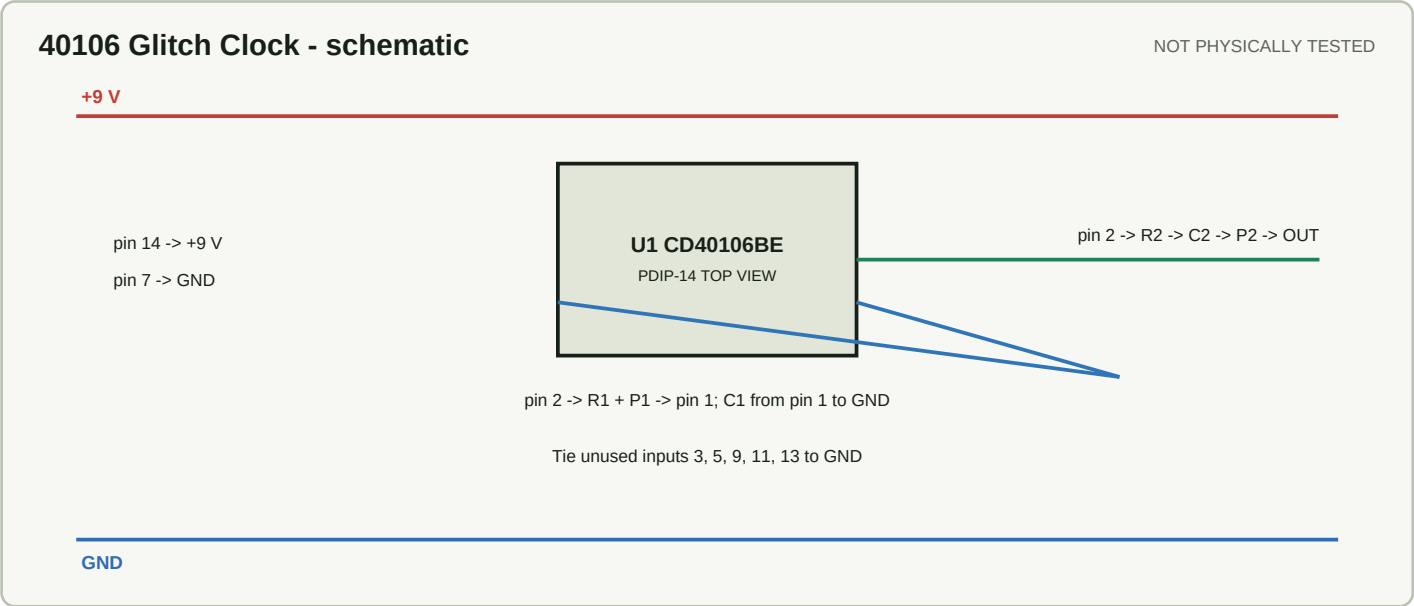
| Ref | Exact component / value                 | Qty |
|-----|-----------------------------------------|-----|
| BB1 | BB830-style solderless breadboard       | 1   |
| U1  | TI CD40106BE PDIP-14                    | 1   |
| P1  | Bourns PTV09 500 kOhm linear            | 1   |
| R1  | 1 kOhm 1/4 W minimum resistor           | 1   |
| R2  | 10 kOhm 1/4 W output resistor           | 1   |
| C1  | 0.1 uF film timing capacitor            | 1   |
| C2  | 1 uF 25 V electrolytic                  | 1   |
| C3  | 100 nF ceramic bypass                   | 1   |
| P2  | Bourns PTV09 100 kOhm linear            | 1   |
| J1  | 6.35 mm mono output jack                | 1   |
| PS1 | Regulated current-limited 9 V DC supply | 1   |

### Pinout verification

CD40106BE package view: TOP VIEW. In the supplied horizontal breadboard layout the notch faces left; DIP numbering proceeds counterclockwise.

| Pin | Name | Role                   |
|-----|------|------------------------|
| 1   | A    | Inverter A input       |
| 2   | G=/A | Inverter A output      |
| 3   | B    | Unused input to ground |
| 4   | H=/B | Unused output          |
| 5   | C    | Unused input to ground |
| 6   | I=/C | Unused output          |
| 7   | VSS  | Ground                 |
| 8   | J=/D | Unused output          |
| 9   | D    | Unused input to ground |
| 10  | K=/E | Unused output          |
| 11  | E    | Unused input to ground |
| 12  | L=/F | Unused output          |
| 13  | F    | Unused input to ground |
| 14  | VDD  | +9 V supply            |

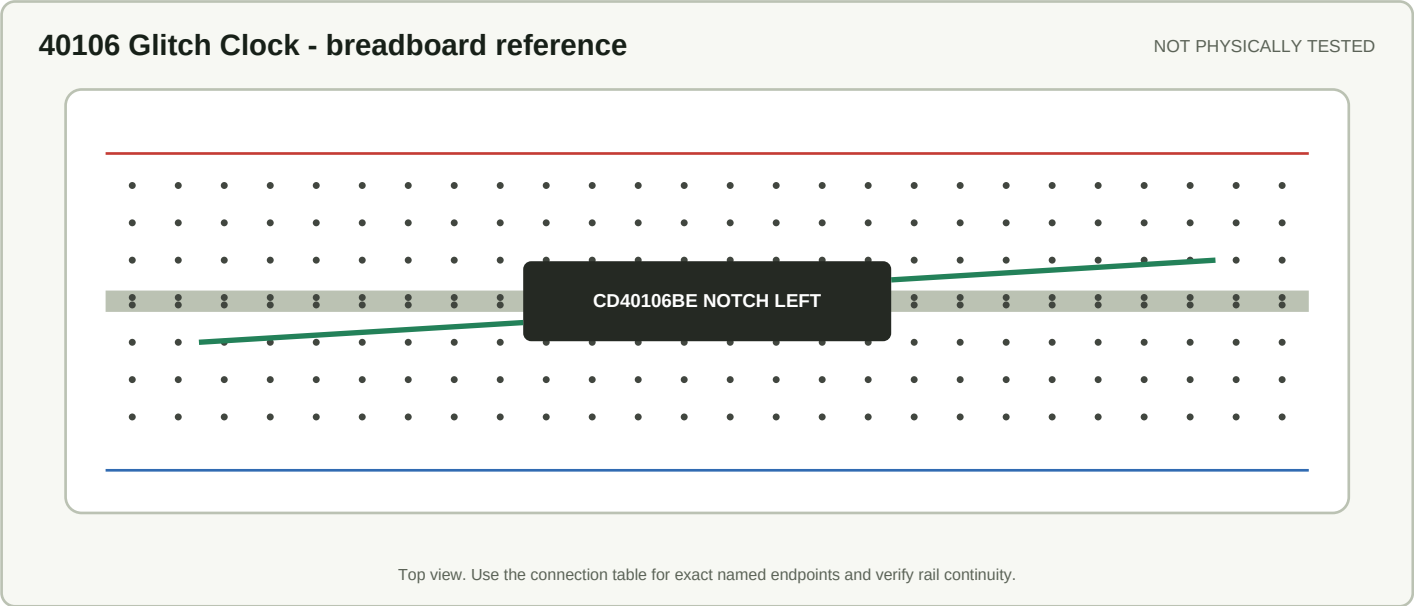
### 3. Labeled schematic



### 4. Canonical connections

| Net    | From                    | To                                   | Wire         |
|--------|-------------------------|--------------------------------------|--------------|
| Power  | +9 V / ground           | U1 pin 14 / pin 7; C3 across supply  | red/black    |
| Clock  | U1 pin 2                | R1 + P1 back to U1 pin 1             | blue         |
| Timing | U1 pin 1                | C1 to ground                         | black        |
| Unused | U1 pins 3, 5, 9, 11, 13 | Ground; leave unused outputs open    | black        |
| Output | U1 pin 2 through R2     | C2 positive; C2 negative to P2 lug 3 | yellow/green |
| Level  | P2 lug 1 to ground      | P2 wiper to J1 tip                   | green        |
| Ground | J1 sleeve               | Shared ground rail                   | black        |

# 5. Breadboard reference



## Breadboard orientation

This drawing is a top-view placement reference. The connection table is authoritative for every electrical net. Typical breadboards connect A-E in each numbered row and F-J separately across the center trench; power-rail continuity varies by product and must be measured with power removed.

## 6. Power-off assembly

**01** Disconnect power, map the rails, and place U1 across the center trench at columns 10-16 with notch left.

**02** Connect U1 pin 14 to +9 V and pin 7 to ground. Put C3 close to those pins.

**03** Connect pin 2 through R1 and P1 back to pin 1. Tie P1 wiper to the used outer lug.

**04** Connect C1 from pin 1 to ground.

**05** Tie unused inputs pins 3, 5, 9, 11, and 13 to ground. Leave unused outputs open.

**06** Connect pin 2 through R2 and then C2 to P2 lug 3. C2 positive faces R2.

**07** Connect P2 lug 1 to ground, the wiper to J1 tip, and J1 sleeve to ground.

**08** Set P1 to high resistance and P2 low. Inspect pins 14 and 7, then apply current-limited 9 V.

## 7. Test procedure and expected results

Start with the supply current limit low and the output level down. Use a multimeter in DC-voltage mode with the black probe on the ground rail. Confirm the power and bias readings before connecting audio. Expected readings below are predictions, not measurements.

| Checkpoint | Calculated / expected observation                       |
|------------|---------------------------------------------------------|
| 1          | U1 pin 14 to pin 7: about 9 V DC                        |
| 2          | U1 pin 1: repeating capacitor ramp                      |
| 3          | U1 pin 2: logic pulse alternating near the supply rails |
| 4          | J1: AC-coupled pulse tone; rate changes with P1         |

## 8. Troubleshooting

1. Remove power. Confirm +9 V and ground are not shorted using resistance or continuity mode.
2. Confirm the IC notch direction and locate pin 1 again. A mirrored package is the most serious common wiring error.
3. Check every polarized capacitor and every diode band against the connection table.
4. Measure the supply at the IC pins. If it is missing, repair power wiring before testing signals.
5. Compare one net at a time with the canonical table. Do not change several wires at once.
6. If DC checks pass but sound is missing, trace from output toward input with a powered audio probe or oscilloscope under instructor supervision.
7. Component tolerances change pitch, cutoff, gain, and bias. A difference from the browser model is not automatically a fault.

## Sources

TI CD40106B: <https://www.ti.com/lit/ds/symlink/cd40106b.pdf>

Bourns PTV09: <https://www.bourns.com/docs/product-datasheets/PTV09.pdf>

### Wire audit statement

The connection table, schematic labels, and breadboard IC orientation were reconciled for this guide. Off-board potentiometers and jacks are identified by terminal name in the table. No physical build or measurement was performed; complete a continuity and power-rail audit before first power-up.